SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1 - ASSESSMENT TOOL 2

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE: CSA06

COURSE OUTCOME COVERED

CO1: Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method.

(BL-1, BL-2, BL-3)

Assessment Tool Weightage:

QUESTIONS: 10 **Total Marks: 100**

Annexure A

Concept Mapping

Code of Conduct:

I P.Sai Ganesh Reddy(192525111) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P.Sai Ganesh Reddy

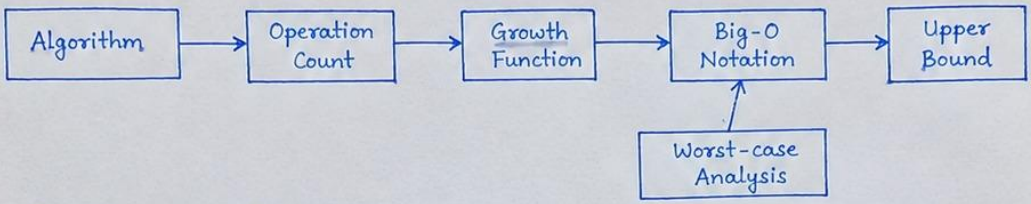
RUBRICS FOR CALCULATION

Criteria	**Weigh tage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

6. Consider the concept map: Algorithm → Operation Count → Growth Function → Big-O Notation → Upper Bound. Map the conceptual links showing how counting basic operations leads to a growth function and then to an asymptotic upper bound. Extend the map by connecting Worst-case Analysis as a supporting node to Big-O Notation. Justify why Big-O is used to represent an upper limit on running time rather than an exact value. Interpret how the completed structure supports reliable performance estimation for large inputs.

Answer:

The completed concept map is shown below:



1. Algorithm → Operation Count :
   - The algorithm is analyzed by counting the number of basic operations (e.g., comparisons, assignments, arithmetic operations).
   - This gives a measure of how much work the algorithm performs.
2. Operation Count → Growth Function :
   - The operation count is expressed as a function of input size n.
   - This function shows how the number of operations grows with n.
3. Growth Function → Big-O Notation :
   - The growth function is represented using asymptotic notation, mainly Big-O.
   - Big-O captures the upper limit of the growth rate of the function.
4. Big-O Notation → Upper Bound :
   - Big-O provides an upper bound on the running time of the algorithm.
   - It shows the maximum rate at which the time can grow for large values of n.
5. Worst-case Analysis (supporting node) → Big-O Notation :
   - The worst-case scenario is used to determine the upper bound.
   - Hence, worst-case analysis supports the use of Big-O notation.

Justification:

Big-O is used to represent an upper limit rather than an exact value because:
- The exact running time depends on many factors (input data, environment, etc.).
- It provides a safe and reliable estimate for all possible inputs.
- An exact value is rarely possible for complex algorithms, especially for large n.

Interpretation:

This concept map shows a clear flow from basic operation counting to a formal asymptotic upper bound. By including worst-case analysis, it ensures that the estimated bound is reliable for large inputs, which helps in predicting and comparing the performance of algorithms effectively.